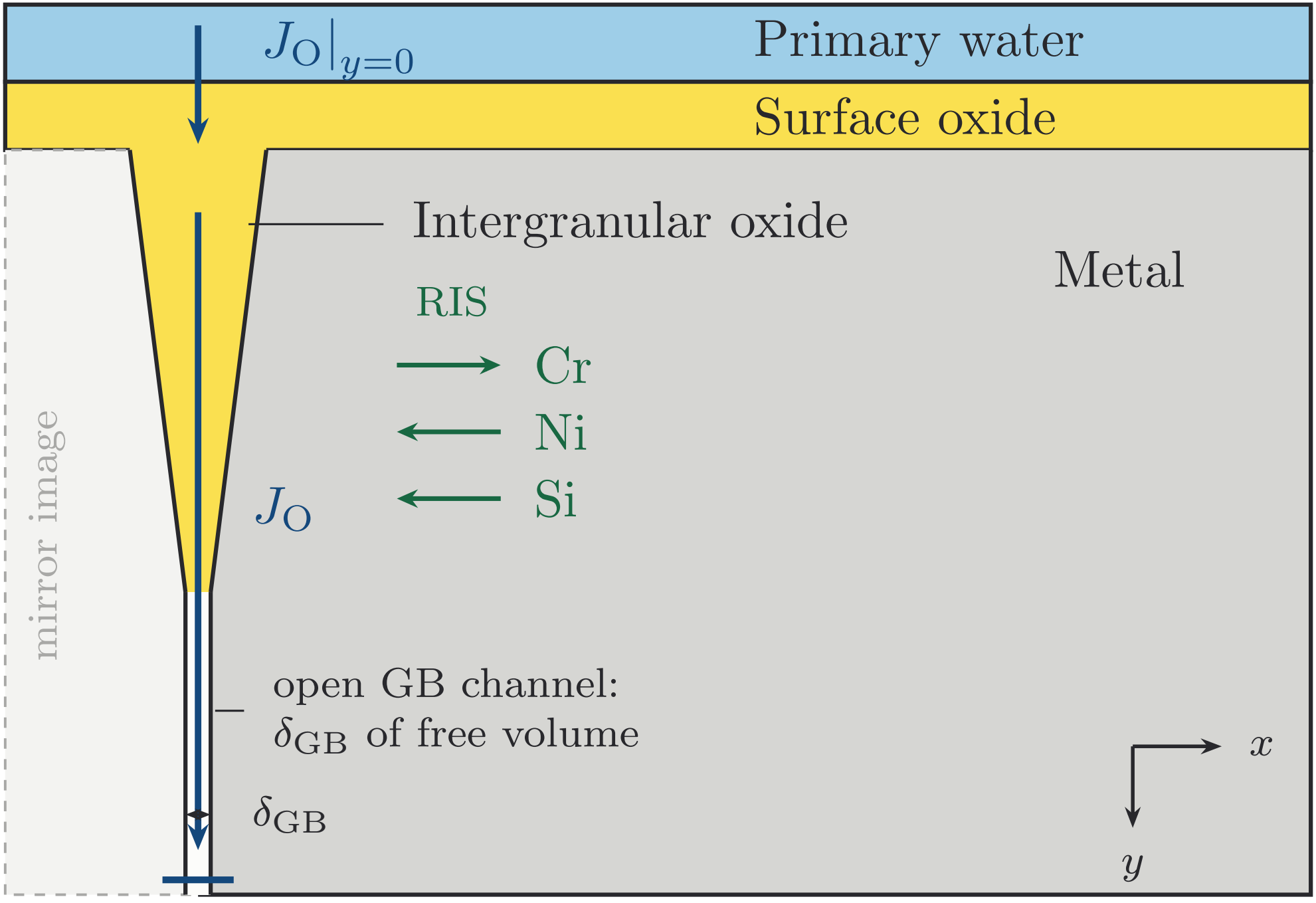

$J_O|_{y=0}$
Primary water
Surface oxide
Intergranular oxide
Metal
RIS
Cr
Ni
Si
J_O
mirror image
open GB channel:
δ_{GB} of free volume
δ_{GB}
x
y